Einsteins Second Postulate: Principle of Relativity	The speed of light in vacuum is the same in all inertial frames of reference and is independent of the motion of the source. → It is impossible for an inertial observer to trav **Time dilation:** $\Delta t = \gamma \Delta t_0$ Proper time between two events (measured in rest frame) Lorentz factor relating the two frames Time interval between same events measured in second frame of reference el at c, the speed of light in vacuum.
Time dilation	**Time dilation:** $\Delta t = \dfrac{\Delta t_0}{\sqrt{1 - u^2/c^2}}$ Proper time between two events (measured in rest frame) Speed of light in vacuum Speed of second frame relative to rest frame Time interval between same events measured in second frame of reference
Lorentz factor	**Lorentz factor** $\gamma = \dfrac{1}{\sqrt{1 - u^2/c^2}}$ Speed of light in vacuum Speed of one frame of reference relative to another
Length contraction	**Length contraction:** $l = l_0\sqrt{1 - \dfrac{u^2}{c^2}} = \dfrac{l_0}{\gamma}$ Proper length of object (measured in rest frame) Speed of second frame relative to rest frame Lorentz factor relating the two frames Length in second frame of reference moving parallel to object's length Speed of light in vacuum
Lorentz coordinate transformation	**Lorentz coordinate transformation:** Spacetime coordinates of an event are x, y, z, t in frame S and x', y', z', t' in frame S'. $x' = \dfrac{x - ut}{\sqrt{1 - u^2/c^2}} = \gamma(x - ut)$ $y' = y$ $z' = z$ $t' = \dfrac{t - ux/c^2}{\sqrt{1 - u^2/c^2}} = \gamma(t - ux/c^2)$ Velocity of S' relative to S in positive direction along x-x'-axis Lorentz factor relating the two frames Speed of light in vacuum
Lorentz velocity transformation	**Lorentz velocity transformation** (velocity in S' in terms of velocity in S): $v'_x = \dfrac{v_x - u}{1 - uv_x/c^2}$ x-velocity of object in frame S' x-velocity of object in frame S Velocity of S' relative to S in positive direction along x-x'-axis Speed of light in vacuum

	Lorentz velocity transformation (velocity in S in terms of velocity in S'): $v_x = \dfrac{v'_x + u}{1 + uv'_x/c^2}$ v_x: x-velocity of object in frame S; v'_x: x-velocity of object in frame S'; u: Velocity of S' relative to S in positive direction along x-x'-axis; c: Speed of light in vacuum
Doppler Effect	**Doppler effect, electromagnetic waves, source approaching observer:** $f = \sqrt{\dfrac{c + u}{c - u}}\, f_0$ f: Frequency measured by observer; f_0: Frequency measured in rest frame of source; u: Speed of source relative to observer; c: Speed of light in vacuum
Relativistic Momentum	**Relativistic momentum** $\vec{p} = \dfrac{m\vec{v}}{\sqrt{1 - v^2/c^2}}$ m: Rest mass of particle; $\vec{v}$: Velocity of particle; v: Speed of particle; c: Speed of light in vacuum
	Relativistic momentum $\vec{p} = \gamma m\vec{v}$ m: Rest mass of particle; $\vec{v}$: Velocity of particle; γ: Lorentz factor relating rest frame of particle and frame of observer
Relativistic Kinetic Energy	**Relativistic kinetic energy** $K = \dfrac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2 = (\gamma - 1)mc^2$ m: Rest mass of particle; c: Speed of light in vacuum; v: Speed of particle; γ: Lorentz factor relating rest frame of particle and frame of observer
Total Energy of a Particle	**Total energy of a particle** $E = K + mc^2 = \dfrac{mc^2}{\sqrt{1 - v^2/c^2}} = \gamma mc^2$ K: Kinetic energy; mc^2: Rest energy; m: Rest mass of particle; c: Speed of light in vacuum; v: Speed of particle; γ: Lorentz factor relating rest frame of particle and frame of observer
Total/ Rest energy and momentunm	**Total energy, rest energy, and momentum:** $E^2 = (mc^2)^2 + (pc)^2$ E: Total energy; mc^2: Rest energy; m: Rest mass; p: Magnitude of momentum; c: Speed of light in vacuum

Key Concepts

KEYCONCEPT The time interval between two events depends on the frame of reference of the observer. The time interval is shortest in an inertial frame S where the two events occur at the same point; this is the proper time Δt_0. In any other inertial frame that is moving with respect to S, the time interval is longer (it is *dilated*) [Eq. (37.6)].

KEYCONCEPT If two events take place at the same point in an inertial frame S, an observer in a frame S' moving relative to S observes a time interval between these events that is greater than in S by the Lorentz factor γ [Eqs. (37.7) and (37.8)]. This factor is substantially different from 1 only if the relative speed of S' and S is a substantial fraction of the speed of light.

KEYCONCEPT No matter which inertial frame of reference you are in, you observe dilated time intervals for timekeepers (clocks) of any kind that move relative to you. If S and S' are two inertial frames in relative motion, an observer in S measures clocks in S' to run slow, and an observer in S' measures clocks in S to run slow.

KEYCONCEPT The dimensions of an object depend on the frame of reference of the observer. If the object is moving at constant velocity relative to your inertial frame, its length is contracted *along* the direction of relative motion [Eq. (37.16)]. There is no change in length *perpendicular* to the direction of relative motion.

KEYCONCEPT No matter which inertial frame of reference you are in, you observe objects that move relative to you to be contracted along the direction of relative motion. If S and S' are two inertial frames in relative motion, an observer in S measures objects in S' to have contracted lengths, and an observer in S' measures objects in S to have contracted lengths.

KEYCONCEPT Given the position and time of an event as measured in one inertial frame of reference, you can use the Lorentz coordinate transformation to find the position and time of that same event as measured in a second inertial frame moving relative to the first one.

KEYCONCEPT Given the velocity of an object as measured in one inertial frame of reference, you can use the Lorentz velocity transformation to find the velocity of that same object as measured in a second inertial frame moving relative to the first one. If an object is moving slower than c as measured in one inertial frame, its speed is less than c as measured in any other inertial frame.

KEYCONCEPT The frequency of electromagnetic waves as measured by an observer moving relative to their source is different from the frequency measured in the rest frame of the source. The measured frequency is greater than the source frequency if the observer and source are approaching, and less than the source frequency if they are receding [Eq. (37.25)].

KEYCONCEPT The expression for momentum [Eq. (37.31)] and the relationship between net force and acceleration [Newton's second law, Eqs. (37.32) and (37.33)] are both modified in the special theory of relativity. As the speed of a material particle approaches c, its momentum approaches infinity and the acceleration caused by a net force approaches zero.

KEYCONCEPT The expression for kinetic energy [Eq. (37.36)] is modified in the special theory of relativity. As the speed of a material particle approaches c, its kinetic energy approaches infinity. The total energy of a particle of rest mass m is the sum of its kinetic energy and its rest energy mc^2.

KEYCONCEPT In a relativistic collision, momentum and total energy are both conserved. Particles can be created or destroyed in such collisions by converting kinetic energy to rest energy, or vice versa.

Cautions

CAUTION **Choose your inertial frame coordinates wisely** Many of the equations derived in this chapter are true *only* if you define your inertial reference frames as stated in the preceding paragraph. The positive x-direction must be the direction in which the origin O' moves relative to the origin O. In Fig. 37.3 this direction is to the right; if instead O' moves to the left relative to O, you must define the positive x-direction to be to the left. When solving problems with reference frames, you should *always* draw a diagram showing which reference frame goes with which observer.

CAUTION **Measuring time intervals** It is important to note that the time interval Δt in Eq. (37.6) involves events that occur *at different space points* in the frame of reference S. Note also that any differences between Δt and the proper time Δt_0 are *not* caused by differences in the times required for light to travel from those space points to an observer at rest in S. We assume that our observer can correct for differences in light transit times, just as an astronomer who's observing the sun understands that an event seen now on earth actually occurred 500 s ago on the sun's surface. Alternatively, we can use *two* observers, one stationary at the location of the first event and the other at the second, each with his or her own clock. We can synchronize these two clocks without difficulty, as long as they are at rest in the same frame of reference. For example, we could send a light pulse simultaneously to the two clocks from a point midway between them. When the pulses arrive, the observers set their clocks to a prearranged time. (But clocks that are synchronized in one frame of reference *are not* in general synchronized in any other frame.)

CAUTION **Length contraction is real** This is *not* an optical illusion! The ruler really is shorter in reference frame S than it is in S'.

CAUTION **Use the correct reference frame coordinates** The Lorentz transformation equations given by Eqs. (37.21), (37.22), and (37.23) assume that frame S' is moving in the positive x-direction with velocity u relative to frame S. Always set up your coordinate system to follow this convention.

CAUTION **Three electron energies** All electrons have *rest* energy 0.511 MeV. An electron accelerated from rest through a 5.00 MeV potential increase has *kinetic* energy 5.00 MeV (we call it a "5.00 MeV electron") and *total* energy 5.51 MeV. Be careful to distinguish these energies from one another.